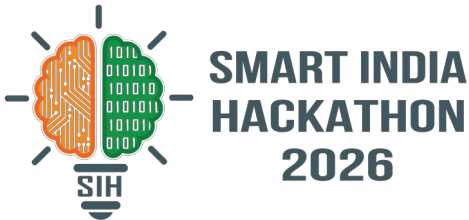


SMART INDIA HACKATHON 2026



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

Problem Statement ID

MDoNER • Smart Automation • Software

PROBLEM STATEMENT ID

SIH26002

PROBLEM STATEMENT TITLE

AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

THEME

Smart Automation

PS CATEGORY

Software

TEAM ID

17 (internal group no.)

TEAM NAME

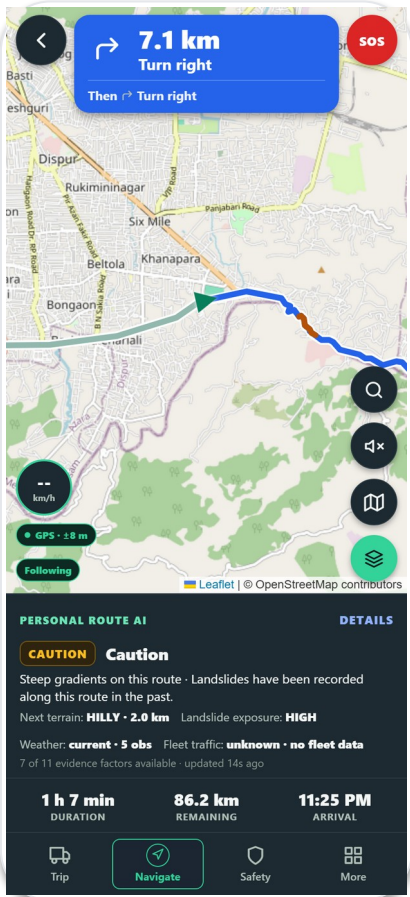
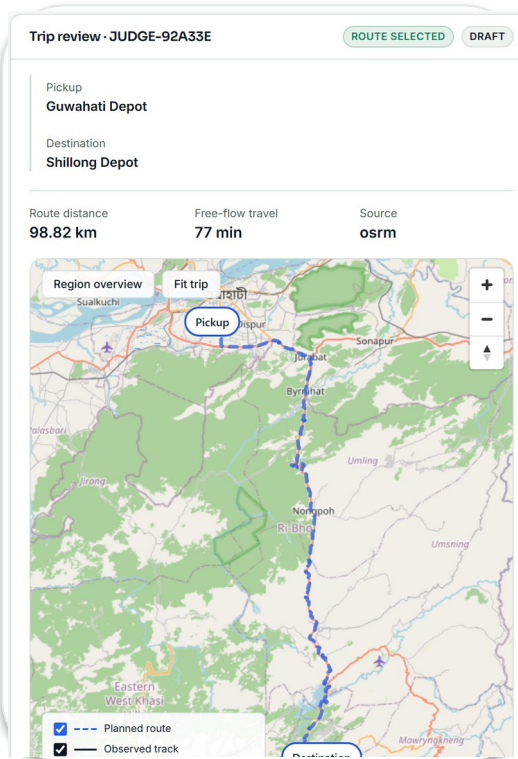
NER-AI LOGISTICS

SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

End-to-end working prototype • physical-phone validated • live demo corridor Guwahati → Shillong

LIVE WORKING SYSTEM manager console + Android driver app



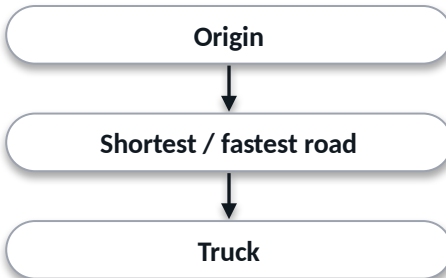
Manager route review • Driver navigation — real screens, real road

PROPOSED SOLUTION

NOT THE SHORTEST ROAD. THE ROAD THAT IS OPERATIONALLY USABLE NOW.

Monsoon · landslides · floods · official warnings · single-lane hill roads · no signal. **A shortest-path router still sends the truck.**

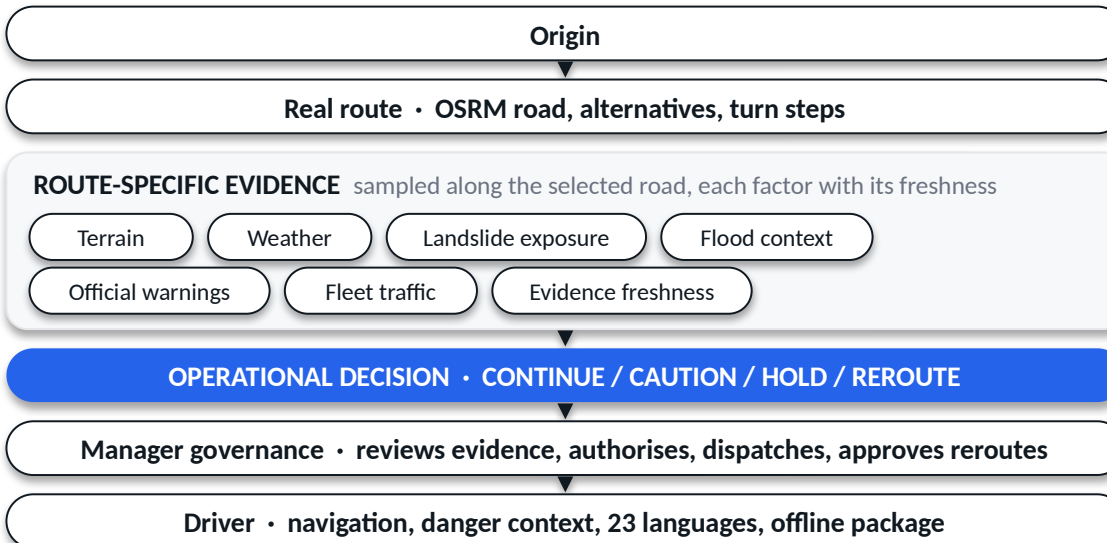
A NORMAL ROUTER



No terrain, weather, hazard or freshness check.

No governance. No answer when data is missing.

RASTA AI evidence before the road



MANAGER · CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN **0.5 KM STEEP**

49–1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY **HIGH EXPOSURE**

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007–2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS

NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umiling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC

UNKNOWN

Real screen · evidence · freshness · UNKNOWN

UNKNOWN ≠ SAFE

The system refuses to fabricate certainty: missing or stale evidence is marked UNKNOWN — never safe — and goes to an authorised reviewer.

TECHNICAL APPROACH

VERIFIED DATA → DECISION → DRIVER provider adapters with health + freshness

WEATHER

Open-Meteo · MET
Norway

TERRAIN

Copernicus DEM ·
OpenTopoData

WARNINGS

NDMA SACHET (CAP)

FLOOD

GloFAS

**LANDSLIDE
HISTORY**

NASA GLC 2007-17

TRAFFIC

fleet GPS fixes

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps · freshness per factor

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE**CAUTION****HOLD****REROUTE****MANAGER REVIEW****DRIVER NAVIGATION****ROUTE-AHEAD MONITOR (60 s)****CONDITIONS CHANGE?****REASSESS / REROUTE**

Re-scored every 60 s on the road ahead. A reroute is proposed to the manager, never applied silently.

BUILT WITH

CLIENTS Manager console (React + TS) · Driver app (React Native / Expo, Android) · one login, server-decided role, offline package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

EVIDENCE Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM

WHO DECIDES AI is supporting

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision and assists in the driver's language — never decides

EXPERIMENTAL ML

research channel only; kept outside routing by the safety-validation gate

GOVERNANCE CHAIN**Verified evidence****Deterministic safety policy****Manager review****Driver action**

NOT A CONCEPT. A WORKING END-TO-END PROTOTYPE.

PLAN

REVIEW

DISPATCH

ACCEPT

TRUCK VERIFY

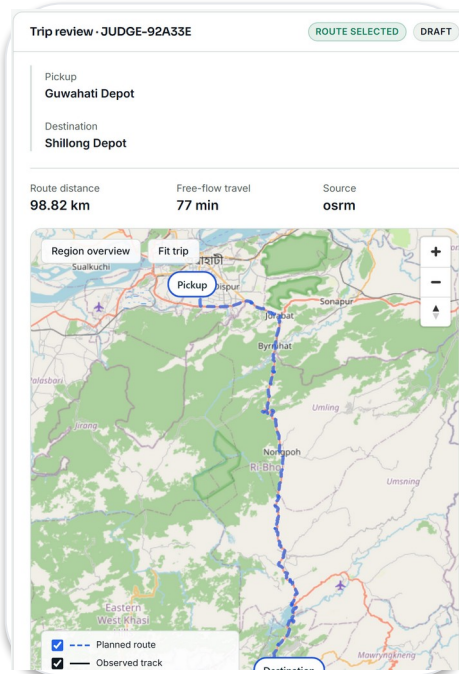
NAVIGATE

OFF-ROUTE

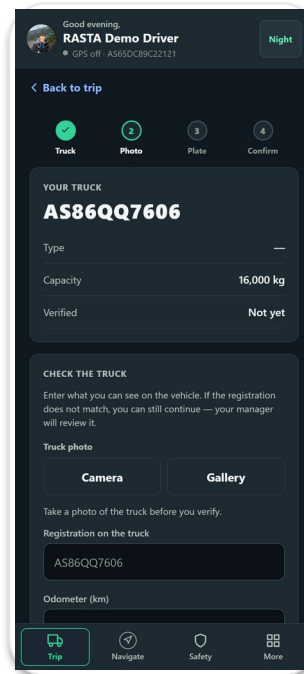
MANAGER REROUTE APPROVAL

DELIVERY

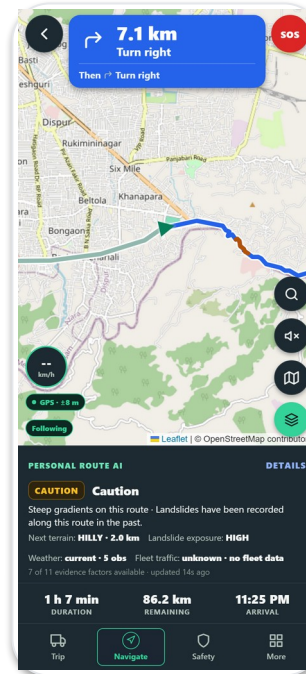
REAL SCREENS FROM THE CERTIFIED RUN



Manager • route review



Driver • truck check



Driver • navigation

PROOF physical Android phone • hosted backend • real road

Run end to end on a physical Android phone, hosted backend, real road.

12/12 physical judge-flow steps**9/9** physical role-flow steps**1138+** backend • **621+** driver • **170+** manager tests**PHYSICAL ANDROID • CERTIFIED****Canonical demo** Guwahati → Shillong • ≈ 98.8 km real route

Every reroute approved by a manager before the phone followed it.

KNOWN LIMITS → MITIGATION Cold start → warm /health first • rate limits → cached evidence + explicit UNKNOWN • push needs Firebase → in-app alerts work • experimental ML → outside routing

IMPACT AND BENEFITS

RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

FOOD • MEDICINE • FUEL • RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

WHO BENEFITS

FLEET MANAGERS

Usability and uncertainty visible before dispatch. Auditable.

DRIVERS

Danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

Resilient access to food, medicine, fuel, relief.

GOVERNMENT / OPS

Auditable evidence and disruption per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

100 % of reroutes reviewed by a human

0 unknown factors counted as safe

HUMAN GOVERNANCE AI has no uncontrolled authority · no invented percentages

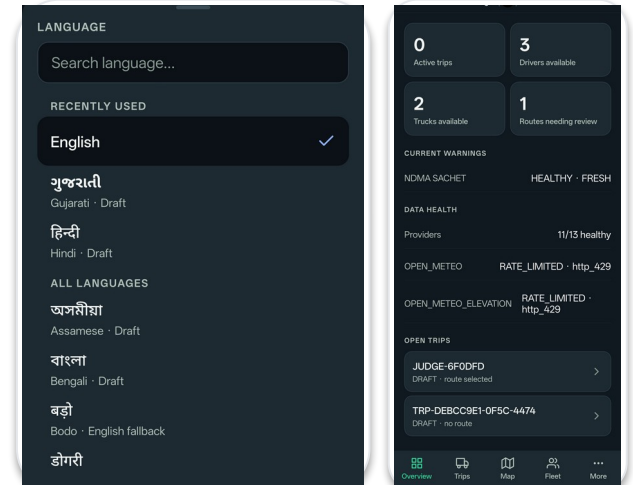
Verified evidence

Deterministic safety policy

Manager governance

Driver action

REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

APK 1.0.18. One login, server-decided role. Every language shows its status — Verified, Draft or English fallback.

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

Road geometry, alternatives, turn steps.

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM •

OpenTopoData

Forecast along the road; climb, gradient, steep segments.

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

Official warnings with freshness; discharge vs 30-day mean.

LANDSLIDE HISTORY

NASA Global Landslide Catalog

Historical exposure, 2007–2017, ≤ 5 km accuracy.

AI ASSISTANCE

Google Gemini

Explains in the driver's language. Never safety authority.

PROBLEM STATEMENT

MDoNER • SIH26002

Smart logistics and accessibility intelligence for the NER.

EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

NER geographic holdout • 154 events • no NER row in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Recall was not enough. Held-out testing exposed the false-positive rate. The gate keeps it out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 — calibrated logistic regression, NASA GLC + ERA5-Land rain + DEM slope

REFERENCES

gpm.nasa.gov/landslides (NASA GLC) • sachet.ndma.gov.in • global-flood.emergency.copernicus.eu (GloFAS) • open-meteo.com • api.met.no • opentopodata.org • Copernicus DEM • project-osrm.org • openstreetmap.org • github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry, error analysis)